

SQL Mini Project

1.

1. create the following tables.

EMP_Details

Column Name	Datatype	Length	Constraint	
Empno	Integer		Primary Key	
First_Name	Varchar2	30	Not Null	
Last_Name	Varchar2	30	Not Null	
Lodge_Name	Varchar2	30	Foreign Key(Lodge_Details)	

EMP_Skill

Note: Empno,Skill are composite Primary Key.

Column Name	Datatype	Length	Constraint	
Empno	Integer		Primary Key Foreign Key(EMP_Details)	
Skill	Varchar2	30	Primary Key Foreign Key(Skill_Details)	
Grade	Number	10		

SKILL_Details

Column Name	Datatype	Length	Constraint	
Skill	Varchar2	30	Primary Key	
Skill_Desc	Varchar2	30		

LODGE_Details

Column Name	Datatype	Length	Constraint	
Lodge_Name	Varchar2	30	Primary Key	
Lodge_Manager	Varchar2	30	Not Null	
Lodge_Address	Varchar2	30		

Output:-

```
SQL> CREATE TABLE LODGE_DETAILS(  
 2     LODGE_NAME VARCHAR2(30) PRIMARY KEY,  
 3     LODGE_MANAGER VARCHAR2(30) NOT NULL,  
 4     LODGE_ADDRESS VARCHAR2(30)  
 5 );
```

Table created.

```
SQL> CREATE TABLE EMP_DETAILS(  
 2     EMPNO NUMBER PRIMARY KEY,  
 3     FIRST_NAME VARCHAR2(30) NOT NULL,  
 4     LAST_NAME VARCHAR2(30) NOT NULL,  
 5     LODGE_NAME VARCHAR2(30),  
 6     FOREIGN KEY (LODGE_NAME)  
 7     REFERENCES LODGE_DETAILS(LODGE_NAME)  
 8 );
```

Table created.

```
SQL> CREATE TABLE SKILL_DETAILS(  
 2     SKILL VARCHAR2(30) PRIMARY KEY,  
 3     SKILL_DESC VARCHAR2(30)  
 4 );
```

Table created.

```
SQL> CREATE TABLE EMP_SKILL(  
 2     EMPNO NUMBER,  
 3     SKILL VARCHAR2(30),  
 4     GRADE NUMBER(10),  
 5     PRIMARY KEY (EMPNO, SKILL),  
 6     FOREIGN KEY (EMPNO)  
 7     REFERENCES EMP_DETAILS(EMPNO),  
 8     FOREIGN KEY (SKILL)  
 9     REFERENCES SKILL_DETAILS(SKILL)  
10 );
```

Table created.

SQL>

2.

2. Insert data into the following tables

LODGE_Details:

Lodge_Name	Lodge_Manager	Lodge_Address
Jessy Lodge	Rajan	Rakesh Nagar
Rajesh Lodge	Rajani	Anna Nagar
Gooday Lodge	John	Sea Road

EMP_Details:

Empno	First_Name	Last_Name	Lodge_Name
101	James	Jackson	Rajesh Lodge
102	Kalpash	Raj	Gooday Lodge
103	Jasmine	Joy	Jessy Lodge

SKILL_Details:

Skill	Skill_Desc
Oracle SQL	Oracle Corporation
Dot Net	Microsoft
Sybase	SAP Corporation

EMP_skill

Empno	Skill	Grade
101	Oracle SQL	9
101	Sybase	8
102	Dot Net	9
103	Oracle SQL	8

```
SQL> INSERT INTO LODGE_DETAILS VALUES
  2  ('Jessy Lodge','Rajan','Rakesh Nagar');
```

```
1 row created.
```

```
SQL>
SQL> INSERT INTO LODGE_DETAILS VALUES
  2  ('Rajesh Lodge','Rajani','Anna Nagar');
```

```
1 row created.
```

```
SQL>
SQL> INSERT INTO LODGE_DETAILS VALUES
  2  ('Gooday Lodge','John','Sea Road');
```

```
1 row created.
```

```
SQL> SELECT * FROM LODGE_DETAILS;
```

LODGE_NAME	LODGE_MANAGER
Jessy Lodge Rakesh Nagar	Rajan
Rajesh Lodge Anna Nagar	Rajani
Gooday Lodge Sea Road	John

```
SQL> INSERT INTO EMP_DETAILS
  2  VALUES(101,'James','Jackson','Rajesh Lodge');
```

```
1 row created.
```

```
SQL>
SQL> INSERT INTO EMP_DETAILS
  2  VALUES(102,'Kalpash','Raj','Gooday Lodge');
```

```
1 row created.
```

```
SQL>
SQL> INSERT INTO EMP_DETAILS
  2  VALUES(103,'John','David','Jessy Lodge');
```

```
1 row created.
```

```
SQL> SELECT * FROM EMP_DETAILS;
```

EMPNO	FIRST_NAME	LAST_NAME
101	James	Jackson
102	Kalpash	Raj
103	John	David

```
SQL> INSERT INTO SKILL_DETAILS VALUES
  2 ('Oracle SQL','Oracle Corporation');
```

1 row created.

```
SQL>
SQL> INSERT INTO SKILL_DETAILS VALUES
  2 ('Dot Net','Microsoft');
```

1 row created.

```
SQL>
SQL> INSERT INTO SKILL_DETAILS VALUES
  2 ('Sybase','SAP Corporation');
```

1 row created.

```
SQL> SELECT * FROM SKILL_DETAILS;
```

SKILL	SKILL_DESC
Oracle SQL	Oracle Corporation
Dot Net	Microsoft
Sybase	SAP Corporation

```
SQL> INSERT INTO EMP_SKILL
  2 VALUES(101,'Oracle SQL',9);
```

1 row created.

```
SQL>
SQL> INSERT INTO EMP_SKILL
  2 VALUES(101,'Sybase',8);
```

1 row created.

```
SQL>
SQL> INSERT INTO EMP_SKILL
  2 VALUES(102,'Dot Net',9);
```

1 row created.

```
SQL>
SQL> INSERT INTO EMP_SKILL
  2 VALUES(103,'Oracle SQL',8);
```

1 row created.

```
SQL> SELECT * FROM EMP_SKILL;
```

EMPNO	SKILL	GRADE
101	Oracle SQL	9
101	Sybase	8
102	Dot Net	9
103	Oracle SQL	8

```
SQL> |
```

3. Employee 101 has changed his lodge to Gooday Lodge.

```
SQL> UPDATE EMP_DETAILS
  2  SET LODGE_NAME = 'Gooday Lodge'
  3  WHERE EMPNO = 101;
```

1 row updated.

```
SQL> SELECT * FROM EMP_DETAILS
  2  WHERE EMPNO = 101;
```

EMPNO	FIRST_NAME	LAST_NAME
101	James	Jackson

Gooday Lodge

SQL> |

4. Employee 103 has resigned from the company and vacated the lodge so delete his details from the respective tables.

```
SQL> DELETE FROM EMP_SKILL
  2  WHERE EMPNO = 103;
```

1 row deleted.

```
SQL> DELETE FROM EMP_DETAILS
  2  WHERE EMPNO = 103;
```

1 row deleted.

```
SQL> SELECT * FROM EMP_DETAILS;
```

EMPNO	FIRST_NAME	LAST_NAME
101	James	Jackson

Gooday Lodge

EMPNO	FIRST_NAME	LAST_NAME
102	Kalpash	Raj

Gooday Lodge

SQL>

```
SQL> SELECT * FROM EMP_SKILL;
```

EMPNO	SKILL	GRADE
101	Oracle SQL	9
101	Sybase	8
102	Dot Net	9

SQL> |

5. The HR Departments want to create reports base on the following requirements

- a. Write a query that will display the employee with skill Sybase. Pass Sybase in any case.
Use Function.
- b. Create a report that will display the First_Name,Skill and Lodge Manager
- c. Create a report that will that display the number of skill that employee 101 has.
- d. Display the employee First_Name who has the Skill Oracle SQL.

A) Write a query that will display the employee with skill Sybase. Pass Sybase in any case.Use Function.

```
SQL> SELECT E.EMPNO,  
2         E.FIRST_NAME,  
3         ES.SKILL  
4 FROM EMP_DETAILS E  
5 JOIN EMP_SKILL ES  
6 ON E.EMPNO = ES.EMPNO  
7 WHERE UPPER(ES.SKILL) = UPPER('Sybase');
```

EMPNO	FIRST_NAME	SKILL
101	James	Sybase

SQL> |

B) Create a report that will display the First_Name,Skill and Lodge Manager.

```
SQL> SELECT E.EMPNO,  
2         E.FIRST_NAME,  
3         ES.SKILL  
4 FROM EMP_DETAILS E  
5 JOIN EMP_SKILL ES  
6 ON E.EMPNO = ES.EMPNO  
7 WHERE UPPER(ES.SKILL) = UPPER('Sybase');
```

EMPNO	FIRST_NAME	SKILL
101	James	Sybase

```
SQL> SELECT E.FIRST_NAME,  
2         ES.SKILL,  
3         L.LODGE_MANAGER  
4 FROM EMP_DETAILS E  
5 JOIN EMP_SKILL ES  
6 ON E.EMPNO = ES.EMPNO  
7 JOIN LODGE_DETAILS L  
8 ON E.LODGE_NAME = L.LODGE_NAME;
```

FIRST_NAME	SKILL
James	Oracle SQL
John	
James	Sybase
John	
Kalpash	Dot Net
John	

SQL> |

C) Create a report that will that display the number of skill that employee 101 has.

```
SQL> SELECT EMPNO,  
2          COUNT(SKILL) AS TOTAL_SKILLS  
3 FROM EMP_SKILL  
4 WHERE EMPNO = 101  
5 GROUP BY EMPNO;
```

EMPNO	TOTAL_SKILLS
101	2

```
SQL>
```

D) Display the employee First_Name who has the Skill Oracle SQL.

```
SQL> SELECT E.FIRST_NAME  
2 FROM EMP_DETAILS E  
3 JOIN EMP_SKILL ES  
4 ON E.EMPNO = ES.EMPNO  
5 WHERE ES.SKILL = 'Oracle SQL';
```

FIRST_NAME
James

James

```
SQL> |
```


6. Create a View EMP_VIEW so that the users can use it that contains columns EMPNO, FIRST_NAME, SKILL, LODGE_MANAGER.

```
SQL> CREATE VIEW EMP_VIEW AS
2  SELECT E.EMPNO,
3         E.FIRST_NAME,
4         ES.SKILL,
5         L.LODGE_MANAGER
6  FROM EMP_DETAILS E
7  JOIN EMP_SKILL ES
8  ON E.EMPNO = ES.EMPNO
9  JOIN LODGE_DETAILS L
10 ON E.LODGE_NAME = L.LODGE_NAME;
```

View created.

```
SQL> SELECT * FROM EMP_VIEW;
```

```

EMPNO FIRST_NAME SKILL
-----
LODGE_MANAGER
-----
John 101 James Oracle SQL
John 101 James Sybase
John 102 Kalpash Dot Net

SQL> |

```

7. Create a Synonym EMPV for the View EMP_VIEW.

```
SQL> CREATE SYNONYM EMPV
2 FOR EMP_VIEW;
```

Synonym created.

```
SQL> SELECT * FROM EMPV;
```

```

EMPNO FIRST_NAME SKILL
-----
LODGE_MANAGER
-----
John      101 James      Oracle SQL
John      101 James      Sybase
John      102 Kalpash    Dot Net

SQL>

```

8. Create Index on Skill column in EMP_Skill table.

```
SQL> CREATE INDEX IDX_SKILL  
2 ON EMP_SKILL(SKILL);
```

Index created.

```
SQL> |
```